



OPEN Multimodal road perception with illumination adaptation in autonomous vehicles

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The development of autonomous driving technology is reshaping transportation methods. However, the significant decline in perception capabilities under low-light conditions such as rain, fog, and nighttime remains a key bottleneck restricting its widespread application. This paper proposes SafeDrive-Fusion, a novel illumination-adaptive multimodal perception framework that introduces dynamic cross-modal attention fusion and safety-critical region prioritization for robust road passability assessment in autonomous driving. Unlike existing static fusion approaches, this framework addresses the problem of significantly reduced perception capabilities in dim environments through an innovative illumination-adaptive weighting mechanism that dynamically integrates visual and radar information while prioritizing safety-critical road regions. Experimental results show that SafeDrive-Fusion maintains 87.5% accuracy even under extremely low-light conditions, with a performance degradation of only 6.1%, far superior to existing methods' degradation range of 16.6%–34.8%. Particularly in safety-critical “hazardous situation” recognition, the system achieves an F1 score of 0.82, which is 0.40 higher than baseline methods. Through its illumination-adaptive modal weight dynamic adjustment mechanism, the system can automatically adjust its dependence on different modalities according to environmental conditions. As illumination decreases from normal to extremely low, the radar modality weight increases from 25% to 62%. Meanwhile, SafeDrive-Fusion features a low computational latency of 45ms, meeting the real-time requirements of autonomous driving.

Background

The development of autonomous driving technology is reshaping transportation methods, yet all-weather safe driving still faces severe challenges^{1,2}. Existing autonomous driving systems perform well under ideal weather and lighting conditions, but show significant performance degradation in complex environments such as rain, fog, snow, and nighttime, limiting their widespread application^{3,4}. The limitations of single perception modalities have become increasingly evident—visual sensors are easily disturbed under low-light and adverse weather conditions, while active sensors like radar, compared to LiDAR, radar offers superior weather penetration capabilities at lower cost, while infrared cameras have limited detection range for highway applications, making radar-visual fusion the optimal choice for practical autonomous driving deployment. although capable of all-weather operation, struggle to provide rich semantic information about the scene^{5,6}. This fluctuation in perception capability directly affects the safety and reliability of driving decisions, becoming a key bottleneck for all-weather deployment of autonomous driving systems^{7,8}.

Road passability status, as a core input for autonomous driving decision-making, requires accurate perception for driving safety⁹. Traditional perception methods mostly focus on object detection and scene segmentation, while research on the overall assessment of road passability status is relatively insufficient¹⁰. Existing research primarily concerns improving the accuracy of perception tasks themselves, with less consideration for how perception results directly support downstream decision-making¹¹. Particularly in adverse weather conditions, how to effectively integrate information from multiple sensors, maintain accurate assessment of the passability status of the road ahead, and transform the assessment results into information that can directly guide driving decisions, has become an urgent scientific problem to solve.

In recent years, lightweight deep learning models and multimodal fusion technologies have made significant progress in the field of autonomous driving, providing new insights for all-weather road passability perception¹². However, most existing multimodal fusion methods adopt simple feature concatenation or weighted fusion strategies, making it difficult to effectively capture the complementary relationships between different modalities, especially lacking dynamic adjustment capabilities when the quality of information from one modality deteriorates¹³. At the same time, the limitations of onboard computing resources place strict

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requirements on the real-time performance of perception algorithms¹⁴. Therefore, designing a computationally efficient multimodal perception system for road passability status that directly supports driving decisions has important theoretical value and practical significance for enhancing the safety and reliability of autonomous driving systems in complex environments.

Existing multimodal fusion approaches in autonomous driving predominantly rely on static weighting strategies that cannot adapt to dynamic environmental changes, particularly the varying reliability of visual sensors under different lighting conditions. These conventional methods employ fixed fusion weights regardless of environmental conditions, failing to prioritize safety-critical information during the fusion process. Furthermore, current approaches lack theoretical frameworks to quantify the relationship between environmental conditions and optimal sensor fusion strategies. Based on these fundamental limitations and the research challenges mentioned above, this paper proposes the SafeDrive-Fusion system, which combines the lightweight and efficient characteristics of EfficientNet with the dynamic adjustment capability of Cross-Attention fusion mechanisms to construct an innovative all-weather multimodal perception framework for road passability status oriented toward autonomous driving. Unlike existing static fusion approaches, SafeDrive-Fusion captures the complementary advantages of vision and radar through dual-modal feature extraction while maintaining perceptual stability under complex weather conditions. Through intelligent illumination-adaptive fusion strategies, it prioritizes safety-critical information and ensures real-time and accurate assessment of road passability status in resource-constrained onboard environments, providing immediately usable decision-making bases for autonomous driving systems. Experimental results show that this method demonstrates excellent robustness and accuracy in all-weather scenarios on the COCO-KITTI dataset, offering an effective solution to the key challenge of all-weather perception for autonomous driving.

Related work

With the rapid development of autonomous driving technology, multimodal perception capability has become a key factor in ensuring system safety and reliability. Jin et al. conducted a systematic review of 3D object detection methods based on LiDAR, cameras, and multi-sensor fusion, emphasizing the advantages of multi-sensor fusion for perception in complex environments¹⁵. The significance of such fusion strategies lies in their ability to effectively compensate for the limitations of single sensors under extreme conditions, especially the degradation of camera visual performance in dim environments. The spatiotemporal calibration method proposed by Lee et al. provides a foundation for precise fusion in multi-camera systems, solving the alignment problem between cameras with different fields of view¹⁶. However, current autonomous driving perception systems still mainly focus on basic object recognition, with insufficient research on the overall assessment of road passability status.

Building upon these foundational approaches, recent research has explored advanced architectures for comprehensive scene understanding. Elhenawy et al. developed vision-language models using CLIP for dynamic scene retrieval, achieving robust performance across diverse weather conditions and demonstrating superior capabilities in complex scenario recognition¹⁷. Guo et al. proposed a multi-task learning approach (YOLO-ODL) for joint detection of traffic objects, drivable areas, and lane lines, achieving state-of-the-art performance on the BDD100K dataset¹⁸. Liu et al. introduced a novel front-to-top view projection framework for monocular bird's-eye view perception, enabling efficient HD map reconstruction from single camera inputs¹⁹. The Vehicle Activity Dataset proposed by Jendoubi et al. represents a significant contribution to multimodal perception research, combining RGB imagery with emission data to enable comprehensive understanding of vehicle-environment interactions²⁰. Despite these advances, research on multimodal perception for road passability assessment in dim environments remains limited, particularly the development of adaptive fusion strategies that can handle varying sensor reliability.

Despite these advances, several challenges remain in autonomous driving perception, particularly in complex and unstructured environments. Real-time segmentation of unstructured road scenes presents significant computational challenges. Bai et al. developed an improved BiSeNet-based model with partial convolution, achieving 78.81 fps while maintaining 72.63% Mean IoU through specialized pooling and attention mechanisms²¹. This approach, however, relies solely on visual information without incorporating complementary sensor modalities. Moreover, it lacks illumination-adaptive mechanisms and does not prioritize safety-critical road regions essential for autonomous driving decisions. Beyond real-time segmentation, anomaly detection represents another critical challenge in autonomous driving perception. For anomaly detection in driving scenes, Wang et al. proposed AnomaskDrive, a mask-based approach that integrates attention mechanisms and refinement strategies, achieving remarkable performance with 98.56% AUC scores on Road Anomaly datasets²². While effective, this method primarily operates on visual data without exploiting radar sensing advantages, employs static attention mechanisms, and lacks dynamic adaptation to illumination conditions. Temporal consistency in video-based perception has been addressed through weakly supervised approaches. Awan and Shin developed multi-class semantic video segmentation techniques leveraging temporal correlations and channel-wise mechanisms for improved road scene understanding²³. This approach remains confined to visual modality processing, lacks real-time performance guarantees, and does not incorporate safety-critical region prioritization. Perspective transformation challenges have been tackled by Liu et al., who demonstrated a front-to-top view projection (FTVP) framework for monocular BEV perception, employing cycle consistency constraints to address distortion issues in traditional view transformation methods¹⁹. This monocular approach suffers from depth perception limitations, lacks robustness under poor visibility conditions, and does not incorporate adaptive fusion strategies for sensor degradation compensation. Behavioral intention prediction has gained attention, as reviewed by Fang et al., who systematically analyzed data-driven approaches for anticipating specific behaviors in driving scenarios²⁴. Current approaches remain limited to basic intention categories, lack adaptation mechanisms for safety-critical scenarios, and do not integrate multimodal sensor information to enhance prediction reliability under varying environmental conditions.

To provide a comprehensive comparison of existing approaches and their limitations, Table 1 summarizes representative work in autonomous driving perception. Current research predominantly employs static fusion strategies that fail to adapt to varying sensor reliability under different environmental conditions, while lacking adequate integration of safety-critical information prioritization and comprehensive frameworks specifically designed for road passability assessment under challenging lighting conditions.

These fundamental limitations highlight the critical need for illumination-adaptive multimodal fusion frameworks that can intelligently balance radar and visual information based on environmental conditions, which represents the core innovation of the SafeDrive-Fusion algorithm proposed in this paper.

Our contributions

This research addresses the challenges of road passability status perception for autonomous driving systems in dim environments by proposing the SafeDrive-Fusion algorithm framework, with the following main contributions:

- **Novel Illumination-Adaptive Multimodal Perception Framework:** Developed an innovative perception framework featuring illumination-adaptive cross-modal attention fusion and safety-critical region masking mechanisms, fundamentally different from existing static fusion approaches. The framework introduces a dynamic environmental responsiveness capability that automatically prioritizes radar information (weight increasing from 25% to 62%) when visual sensors are compromised, while maintaining focus on safety-critical road regions through specialized attention masking. This approach achieves 87.5% accuracy under extremely low-light conditions, representing a 34.7 percentage point improvement over conventional methods, while preserving real-time performance (45ms latency) essential for autonomous driving applications.
- **Dynamic Cross-Modal Fusion Strategy with Environmental Adaptation:** Implemented a pioneering illumination-adaptive fusion strategy that transcends traditional fixed-weight multimodal integration by establishing real-time sensor reliability assessment and dynamic weight redistribution mechanisms. Unlike existing approaches that employ static fusion weights regardless of environmental conditions, this strategy continuously monitors lighting conditions and intelligently adjusts visual-radar contribution balance, demonstrating exceptional effectiveness in safety-critical scenario recognition with an F1 score of 0.82 for hazardous road conditions—a 0.40 improvement over state-of-the-art methods. This dynamic adaptation capability addresses a fundamental limitation in current autonomous driving perception systems.
- **Theoretical Foundation for Adaptive Multimodal Fusion:** Established a comprehensive mathematical framework that quantifies the relationship between environmental lighting conditions and multimodal fusion gains, providing theoretical justification for adaptive sensor integration strategies. This framework introduces novel theorems demonstrating how fusion benefits grow exponentially as environmental conditions deteriorate, validated through extensive experimentation showing only 6.1% performance degradation under extremely low-light conditions compared to 16.6%–34.8% degradation in comparison methods. The theoretical foundation enables predictive fusion weight optimization and provides guidelines for designing robust multimodal perception systems in challenging environments.

Authors	Application scenario	Research content	Potential limitations
Jendoubi et al. ²⁰	Multimodal dataset for eco-routing	Vehicle Activity Dataset combining emission data, RGB imagery, and object detection for comprehensive road scene analysis	Focuses on emission analysis rather than real-time perception, lacks consideration of varying environmental conditions and adaptive fusion strategies
Elhenawy et al. ¹⁷	Dynamic scene understanding	CLIP-based vision-language models for autonomous driving scene classification achieving 91.1% F1-score with real-time edge deployment capability	Limited to scene classification tasks, does not address multimodal sensor fusion or illumination adaptation challenges
Guo et al. ¹⁸	Multi-task road scene understanding	YOLO-ODL framework for joint detection of traffic objects, drivable areas, and lane lines using hard parameter sharing strategy	Focuses on visual perception only, lacks integration with complementary sensor modalities and adaptation to varying lighting conditions
Bai et al. ²¹	Real-time unstructured road segmentation	Improved BiSeNet-based model with partial convolution achieving 78.81 fps and 72.63% Mean IoU for complex road scenes	Single-modal approach without considering sensor fusion, limited robustness under adverse weather and lighting conditions
Wang et al. ²²	Anomaly detection in driving scenes	AnomaskDrive with mask-based attention mechanism achieving 98.56% AUC on Road Anomaly datasets for unknown object detection	Primarily visual-based approach, lacks integration with other sensing modalities for enhanced reliability in challenging environments
Awan and Shin ²³	Weakly supervised video segmentation	Multi-class semantic video segmentation leveraging temporal correlations and channel-wise temporal split mechanisms for road scenes	Limited to visual modality, does not exploit complementary information from other sensors, lacks real-time performance guarantees
Liu et al. ¹⁹	Monocular BEV perception	Front-to-top view projection framework addressing distortion issues in perspective transformation for HD map reconstruction	Monocular vision-based approach with inherent limitations in depth perception and robustness under poor visibility conditions
Cao et al. ²⁴	Autonomous driving object detection	YOLOv7-based object detection with attention mechanisms, achieving improved performance through optimized backbone networks	Single visual modality focus, lacks exploration of multimodal fusion strategies and adaptation to environmental variations
Bao et al. ²⁵	Binocular depth estimation	Self-supervised stereo depth estimation with correction mechanisms for improved accuracy in autonomous driving applications	Relies primarily on visual information, vulnerable to performance degradation in low-contrast and dim lighting environments

Table 1. Comparison of Related Work in Autonomous Driving Perception.

Method

Problem statement

During operation, autonomous vehicles need to process information obtained from sensors such as cameras and radar in real-time to evaluate the passability status of the road ahead. Consider that at time t , the raw multimodal data acquired by the vehicle can be represented as:

$$\mathcal{D}_t = \{I_t, R_t\} \quad (1)$$

where $I_t \in \mathbb{R}^{H \times W \times 3}$ represents RGB image data, with H and W being the image height and width; $R_t \in \mathbb{R}^{H' \times W' \times C}$ represents radar data, with H' and W' being the spatial dimensions of the radar data, and C being the number of channels in the radar data.

The research specifically focuses on the impact of dim environmental conditions on sensor data quality, which can be described by introducing an illumination intensity function l_t :

$$l_t = f_{light}(h_t, w_t, a_t) \quad (2)$$

where h_t represents the time of day (such as morning, dusk, late night, etc.), w_t represents weather conditions (such as cloudy, rainy, etc. that further reduce illumination), and a_t represents the light absorption rate of the surrounding environment (such as tunnels, shaded areas, etc.). The value range of l_t is $[0, 1]$, with smaller values indicating dimmer environments.

Considering the impact of dim environments on sensor data, the actual sensor data obtained can be represented as:

$$\tilde{I}_t = I_t \cdot g_I(l_t) + n_I(l_t) \quad (3)$$

$$\tilde{R}_t = R_t \cdot g_R(l_t) + n_R(l_t) \quad (4)$$

where $g_I(l_t)$ and $g_R(l_t)$ represent the attenuation functions of dim environments on visual and radar sensors respectively, and $n_I(l_t)$ and $n_R(l_t)$ represent the noise introduced by dim environments. Importantly, $g_I(l_t)$ is typically more sensitive to lighting conditions than $g_R(l_t)$, meaning that the performance of visual sensors degrades more significantly in dim environments.

Road passability status can be categorized into four classes: clear passage (c_1), partially blocked (c_2), completely blocked (c_3), and hazardous situation (c_4). The road passability status classification function is defined as:

$$y_t = \Phi(\tilde{I}_t, \tilde{R}_t) \quad (5)$$

where $y_t \in \{c_1, c_2, c_3, c_4\}$ is the label for road passability status, and Φ is an ideal classification function that can accurately determine road passability status under any lighting conditions.

Autonomous driving decisions depend on precise judgments of road passability status. The decision utility function is defined as:

$$U_t = h(y_t, a_t) \quad (6)$$

where a_t represents the action taken by the autonomous driving system based on perception results, and h is a utility function that measures the match between decisions and actual road conditions. When y_t is consistent with the actual road conditions, U_t reaches its maximum value; when inconsistent, U_t decreases, and the greater the degree of inconsistency, the more U_t decreases.

Based on the above mathematical description, the problem of multimodal perception of road passability status in dim environments can be summarized as:

Problem 1 Given multimodal data $\{\tilde{I}_t, \tilde{R}_t\}$ affected by dim environment l_t , how to design a robust classification function $\hat{\Phi}$ to maximize classification accuracy under various lighting conditions, especially in dim environments:

$$\max_{\hat{\Phi}} \mathbb{E}_{l_t} [\mathbb{I}(\hat{\Phi}(\tilde{I}_t, \tilde{R}_t) = \Phi(I_t, R_t))] \quad (7)$$

where $\mathbb{I}(\cdot)$ is an indicator function that takes the value 1 when the condition in the parentheses holds and 0 otherwise; $\mathbb{E}_{l_t}$ represents the expectation over various lighting conditions, with special emphasis on the expected value under dim conditions. At the same time, $\hat{\Phi}$ needs to satisfy the real-time constraints of the onboard computing environment and provide direct support for autonomous driving decisions.

EfficientNet: efficient feature extraction

Introduction to EfficientNet algorithm

- **Limitations of Traditional Methods:** Traditional deep neural networks such as ResNet and VGG typically rely on increasing network depth or width to improve performance, leading to a sharp increase in parameter count and computational complexity²⁶. This scaling approach ignores the balance relationship between depth, width, and resolution, making real-time processing difficult in resource-constrained onboard environments, and prone to insufficient feature representation capability under low-light conditions²⁷.

- **Advantages of EfficientNet:** EfficientNet optimizes network depth, width, and input resolution simultaneously through Compound Scaling, establishing a mathematical relationship between these factors. This method achieves higher quality feature extraction with fewer parameters and further improves computational efficiency through MBConv blocks (Mobile Inverted Bottleneck Convolution). These characteristics make EfficientNet particularly suitable for applications in resource-constrained scenarios such as autonomous driving, providing high-quality feature representations while ensuring real-time performance, as shown in Fig. 1.

EfficientNet dual-modal feature extraction

To address the challenges of road passability status perception in dim environments, EfficientNet optimizes the balance of network depth, width, and resolution through compound scaling to achieve efficient feature extraction in resource-constrained environments. The following provides a detailed mathematical description of this module. Given the input data $\tilde{I}_t$ and $\tilde{R}_t$ at time t , the EfficientNet dual-modal feature extraction process can be formalized as:

$$F_t^I = \mathcal{E}_I(\tilde{I}_t; \theta_I) \in \mathbb{R}^{h \times w \times d_I} \quad (8)$$

$$F_t^R = \mathcal{E}_R(\tilde{R}_t; \theta_R) \in \mathbb{R}^{h \times w \times d_R} \quad (9)$$

where $\mathcal{E}_I$ and $\mathcal{E}_R$ represent the EfficientNet feature extractors for visual and radar modalities respectively, θ_I and θ_R are the corresponding network parameters, h and w are the height and width of the feature map, and d_I and d_R are the feature dimensions. Considering the significant impact of dim environments on visual information, an illumination-adaptive feature enhancement mechanism is introduced to enhance model robustness under different lighting conditions:

$$\hat{F}_t^I = F_t^I \odot (1 + \alpha \cdot \mathcal{A}(F_t^I; \phi_I)) \cdot \beta(l_t) \quad (10)$$

where $\odot$ represents element-wise multiplication, $\mathcal{A}(\cdot; \phi_I)$ is a channel attention function with parameters ϕ_I , α is an enhancement intensity coefficient, and $\beta(l_t)$ is an illumination compensation function:

$$\beta(l_t) = \frac{\gamma}{1 + \exp(-\eta \cdot (l_{th} - l_t))} + 1 \quad (11)$$

where l_{th} is an illumination threshold, and γ and η are adjustment parameters. When $l_t < l_{th}$ (dim environment), $\beta(l_t)$ provides stronger feature enhancement; when $l_t \geq l_{th}$, the enhancement effect gradually weakens. To capture semantic information at different spatial scales, multi-scale feature aggregation is applied:

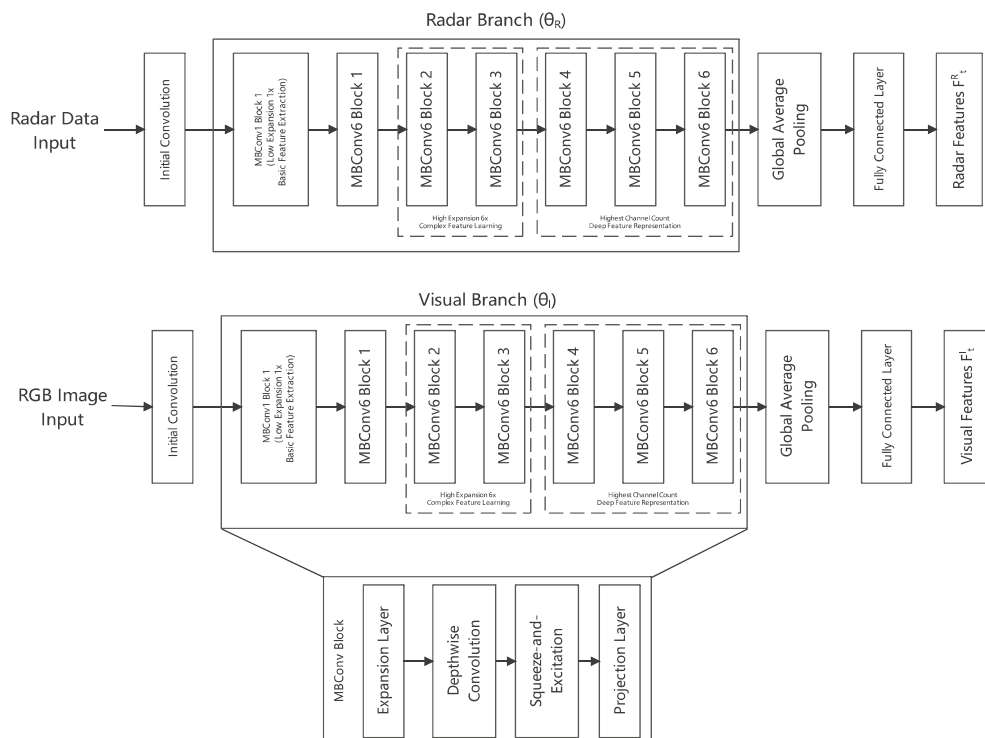


Fig. 1. EfficientNet Algorithm.

$$M_t^I = \sum_{s \in \mathcal{S}} w_s \cdot \mathcal{F}_s(\hat{F}_t^I; \psi_s^I) \quad (12)$$

$$M_t^R = \sum_{s \in \mathcal{S}} w_s \cdot \mathcal{F}_s(F_t^R; \psi_s^R) \quad (13)$$

where $\mathcal{S}$ represents the set of scales considered, $\mathcal{F}_s(\cdot; \psi_s)$ represents the feature transformation function at scale s with corresponding parameters ψ_s , and w_s is the scale weight satisfying $\sum_{s \in \mathcal{S}} w_s = 1$. Finally, to ensure feature representation consistency and complementarity, a feature unified mapping function is designed:

$$Z_t^I = \mathcal{P}_I(M_t^I; \omega_I) \in \mathbb{R}^{h \times w \times d}, \quad Z_t^R = \mathcal{P}_R(M_t^R; \omega_R) \in \mathbb{R}^{h \times w \times d} \quad (14)$$

where $\mathcal{P}_I$ and $\mathcal{P}_R$ are feature mapping functions with parameters ω_I and ω_R , and d is the unified feature dimension.

Based on the above mathematical description, the following theorem can be derived (detailed proof provided in Appendix??):

Theorem 1 (Dual-Modal Feature Complementarity) For a given illumination condition l_t and feature dimension d , there exists an optimal parameter set $\Theta^* = \{\theta_I^*, \theta_R^*, \phi_I^*, \psi_s^{I*}, \psi_s^{R*}, \omega_I^*, \omega_R^*\}$ that maximizes the mutual information between visual and radar features while maintaining the discriminative ability of each modality:

$$\Theta^* = \arg \max_{\Theta} \lambda_1 \cdot MI(Z_t^I, Z_t^R) + \lambda_2 \cdot (I(Z_t^I, y_t) + I(Z_t^R, y_t)) \quad (15)$$

where $MI(\cdot, \cdot)$ represents mutual information, $I(\cdot, \cdot)$ represents information quantity, y_t is the road passability status label, and λ_1 and λ_2 are balancing coefficients.

From this theorem, the following corollary can be derived (proof in Appendix??):

Corollary 1 (Dim Environment Adaptability) When the illumination condition l_t decreases below the threshold l_{th} , there exists $\delta > 0$ such that the features Z_t^I and Z_t^R obtained through EfficientNet dual-modal feature extraction satisfy:

$$\frac{I(Z_t^I \oplus Z_t^R, y_t)}{I(Z_t^I, y_t)} \geq 1 + \delta \cdot (l_{th} - l_t) \quad (16)$$

where $\oplus$ represents feature concatenation operation, and δ is a gain coefficient, indicating that as the environment darkens, the information gain brought by multimodal fusion grows exponentially.

Cross-attention fusion: critical information integration

Introduction to cross-attention fusion algorithm

- **Limitations of Traditional Fusion Methods:** Traditional multimodal fusion methods such as feature concatenation, weighted averaging, and gating mechanism-based fusion typically adopt fixed weight strategies or simple adaptive mechanisms, making it difficult to fully capture the complementary relationships between different modalities²⁸. These methods lack dynamic adjustment capabilities when the quality of information from one modality deteriorates (such as the visual modality in dim environments) and struggle to identify and prioritize safety-critical information, resulting in inefficient fusion²⁹.
- **Advantages of Cross-Attention Fusion:** The Cross-Attention Fusion mechanism establishes attention mappings between modalities, enabling dynamic interaction and selective fusion of cross-modal information. This mechanism uses the features of one modality as queries (Query) and the features of another modality as keys (Key) and values (Value), achieving precise extraction of complementary information through attention weight calculation. Combined with an illumination-adaptive modulation function and safety-critical region masks, the system can automatically adjust its dependence on different modalities based on environmental conditions and prioritize spatial regions directly related to driving safety, significantly improving perception robustness in dim environments, as shown in Fig. 2.

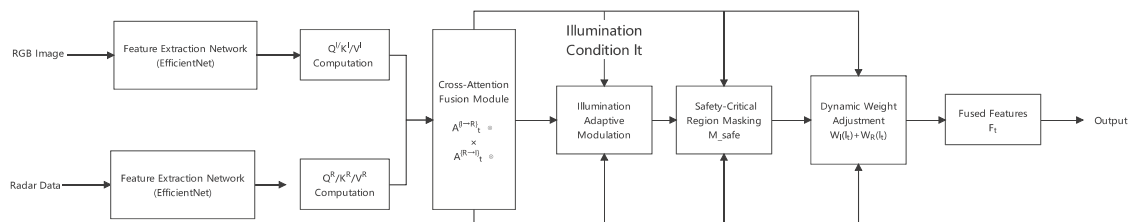


Fig. 2. Cross-Attention Fusion Algorithm.

The selection of radar as the complementary sensing modality in SafeDrive-Fusion is based on comprehensive analysis of practical deployment requirements and sensing characteristics. Compared to LiDAR systems, millimeter-wave radar demonstrates superior penetration capabilities through adverse weather conditions such as rain and fog, while maintaining significantly lower cost (approximately \$1,000 versus over \$10,000 for automotive-grade LiDAR) and power consumption, making it more suitable for widespread commercial deployment. Although infrared cameras offer enhanced performance under low-light conditions, their limited detection range (typically less than 50 meters) and susceptibility to temperature variations restrict their applicability for highway-speed autonomous driving scenarios that require long-range perception capabilities. The radar-visual fusion approach leverages radar's weather immunity and extended detection range while compensating for its limited semantic richness through visual information, achieving an optimal balance between performance, reliability, and cost-effectiveness for practical autonomous driving applications.

Cross-attention-based critical information fusion

The Cross-Attention-based critical information fusion mechanism can solve the problem of road passability status perception in dim environments. This mechanism dynamically evaluates the reliability of features from different modalities, focusing on information directly related to road safety, thereby improving the model's inference capability in dim environments. Given the visual features $Z_t^I \in \mathbb{R}^{h \times w \times d}$ and radar features $Z_t^R \in \mathbb{R}^{h \times w \times d}$ output by the EfficientNet dual-modal feature extraction module, the Cross-Attention fusion process first reorganizes spatial features into sequence form:

$$Q_t^I = \mathcal{T}_Q(Z_t^I; \Omega_Q^I) \in \mathbb{R}^{N \times d_k}, \quad K_t^R = \mathcal{T}_K(Z_t^R; \Omega_K^R) \in \mathbb{R}^{N \times d_k}, \quad V_t^R = \mathcal{T}_V(Z_t^R; \Omega_V^R) \in \mathbb{R}^{N \times d_v} \quad (17)$$

$$Q_t^R = \mathcal{T}_Q(Z_t^R; \Omega_Q^R) \in \mathbb{R}^{N \times d_k}, \quad K_t^I = \mathcal{T}_K(Z_t^I; \Omega_K^I) \in \mathbb{R}^{N \times d_k}, \quad V_t^I = \mathcal{T}_V(Z_t^I; \Omega_V^I) \in \mathbb{R}^{N \times d_v} \quad (18)$$

where $\mathcal{T}_Q$, $\mathcal{T}_K$, and $\mathcal{T}_V$ are transformation functions for query, key, and value respectively, Ω represents the corresponding parameters, $N = h \times w$ is the total number of spatial positions, and d_k and d_v are the dimensions of keys and values respectively. Considering the impact of dim environments on the reliability of the visual modality, an illumination-adaptive attention mechanism is introduced:

$$A_t^{I \rightarrow R} = \text{softmax} \left(\frac{Q_t^I (K_t^R)^T}{\sqrt{d_k}} \cdot \xi(l_t, P_t^I, P_t^R) \right) \in \mathbb{R}^{N \times N} \quad (19)$$

$$A_t^{R \rightarrow I} = \text{softmax} \left(\frac{Q_t^R (K_t^I)^T}{\sqrt{d_k}} \cdot \xi(l_t, P_t^R, P_t^I) \right) \in \mathbb{R}^{N \times N} \quad (20)$$

where $\xi(l_t, P_t^A, P_t^B)$ is an illumination-adaptive modulation function:

$$\xi(l_t, P_t^A, P_t^B) = \exp \left(\kappa \cdot \frac{(1 - l_t) \cdot \mathcal{G}(P_t^A, P_t^B)}{\tau + \mathcal{R}(P_t^A)} \right) \cdot M_{safe} \quad (21)$$

where P_t^A and P_t^B are feature reliability estimates for modalities A and B, $\mathcal{G}(\cdot, \cdot)$ is a complementarity metric function, $\mathcal{R}(\cdot)$ is a reliability evaluation function, κ and τ are adjustment parameters, and $M_{safe} \in \mathbb{R}^{N \times N}$ is a safety-critical region attention mask used to highlight spatial regions directly related to driving safety. Based on the calculated cross-modal attention matrices, the fused feature representations are:

$$F_t^{I \leftarrow R} = A_t^{I \rightarrow R} V_t^R + \lambda_I \cdot Z_t^I, \quad F_t^{R \leftarrow I} = A_t^{R \rightarrow I} V_t^I + \lambda_R \cdot Z_t^R \quad (22)$$

where λ_I and λ_R are residual connection coefficients used to retain original modality information. To achieve prioritized extraction of safety-critical information in dim environments, an adaptive fusion function is designed:

$$F_t = \sigma \left(\mathcal{W}_I(l_t) \cdot F_t^{I \leftarrow R} + \mathcal{W}_R(l_t) \cdot F_t^{R \leftarrow I} \right) \odot \left([F_t^{I \leftarrow R}, F_t^{R \leftarrow I}] ; \Psi \right) \quad (23)$$

where $\mathcal{W}_I(l_t)$ and $\mathcal{W}_R(l_t)$ are illumination-dependent modality weight functions, satisfying $\mathcal{W}_I(l_t) + \mathcal{W}_R(l_t) = 1$ and $\mathcal{W}_R(l_t)$ increases as l_t decreases; σ is a non-linear activation function; $\mathcal{G}$ is a gating mechanism with parameters Ψ ; $[\cdot, \cdot]$ represents channel-dimension concatenation; and $\odot$ is element-wise multiplication. Based on the above fusion mechanism, the following theorem can be derived (detailed proof provided in Appendix??):

Theorem 2 (Modal Adaptability in Dim Environments) For any illumination condition $l_t \in [0, 1]$, there exists an optimal parameter set $\Phi^* = \{\Omega^*, \kappa^*, \tau^*, \lambda_I^*, \lambda_R^*, \Psi^*\}$ that minimizes the expected loss of fused features F_t on the road passability status classification task:

$$\Phi^* = \arg \min_{\Phi} \mathbb{E}_{l_t} \left[\mathcal{L}(f(F_t; \Gamma), y_t) + \rho \cdot \text{KL}(p(y_t | F_t), p(y_t | Z_t^I \oplus Z_t^R)) \right] \quad (24)$$

where $f(\cdot; \Gamma)$ is a classifier, $\mathcal{L}$ is a classification loss function, $\text{KL}(\cdot, \cdot)$ is the KL divergence, ρ is a regularization coefficient, and $\oplus$ represents feature concatenation operation.

From this theorem, the following corollary can be derived (proof in Appendix??):

Corollary 2 (Safety-Critical Information Perception Gain) Under dim environment conditions $l_t < l_{th}$, the Cross-Attention-based safety-critical information fusion method, compared to simple feature concatenation method, shows perception accuracy improvement in safety-critical regions that is negatively correlated with the degree of illumination reduction:

$$\Delta Acc_{safe}(l_t) = Acc_{CA}(l_t) - Acc_{concat}(l_t) \geq \nu \cdot (l_{th} - l_t)^2 \quad (25)$$

where Acc_{CA} and Acc_{concat} represent the classification accuracy in safety-critical regions using Cross-Attention fusion and simple feature concatenation methods respectively, and ν is a proportionality coefficient.

Introduction to SafeDrive-fusion algorithm

To address the challenges of multimodal perception in dim environments, SafeDrive-Fusion integrates illumination-adaptive cross-modal attention fusion with efficient feature extraction for robust road passability assessment. Figure 3 illustrates the complete system architecture, showing the end-to-end pipeline from dual-modal input processing to road passability classification.

The framework operates through three coordinated stages: (1) dual-branch EfficientNet feature extraction with independent parameters (θ_I, θ_R) for visual and radar modalities, (2) illumination-adaptive cross-attention fusion that dynamically adjusts modal weights based on environmental conditions, and (3) safety-critical region prioritization that ensures driving-relevant information receives appropriate attention. The illumination condition l_t serves as a global modulation signal, enabling the system to adapt its fusion strategy in real-time according to lighting variations.

The detailed algorithmic implementation of this architecture is presented in Algorithm 1, which formalizes the mathematical operations described in the preceding sections.

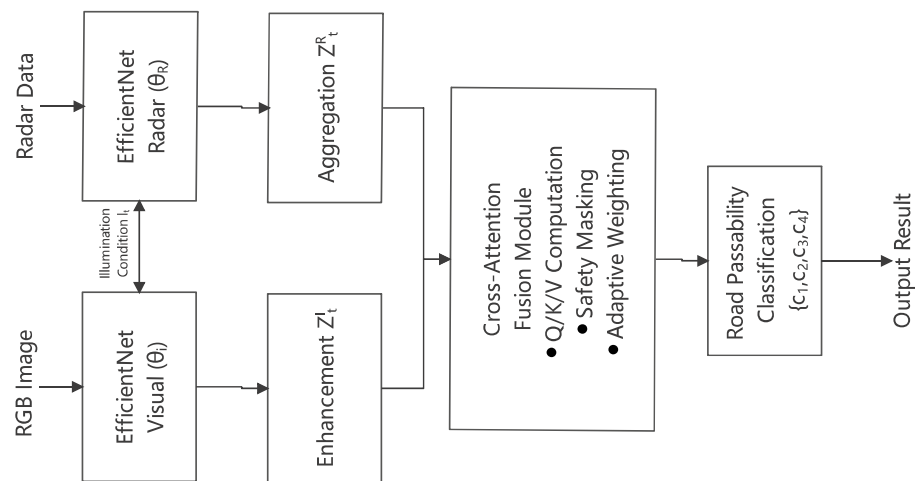


Fig. 3. SafeDrive-Fusion Overall Architecture: An illumination-adaptive multimodal perception framework featuring dual-branch feature extraction, cross-modal attention fusion, and safety-critical region prioritization for robust road passability assessment.

Input: RGB image $\tilde{I}_t$, radar data $\tilde{R}_t$, illumination condition I_t
Output: Road passability status prediction $\hat{y}_t \in \{c_1, c_2, c_3, c_4\}$
Parameter initialization: $\Theta = \{\theta_I, \theta_R, \phi_I, \psi_s^I, \psi_s^R, \omega_I, \omega_R\}$, $\Phi = \{\Omega, \kappa, \tau, \lambda_I, \lambda_R, \Psi\}$, Γ ;

```

/* Stage 1: EfficientNet Dual-Modal Feature Extraction and Enhancement */
Extract visual features  $\mathbf{F}_t^I$  and radar features  $\mathbf{F}_t^R$  using equations 8 and 9 respectively;
Calculate illumination compensation coefficient  $\beta(I_t)$  based on equation 11;
Perform adaptive enhancement of visual features according to equation 10:  $\hat{\mathbf{F}}_t^I$ ;

/* Multi-scale Feature Processing */
For all scales  $s \in \mathcal{S}$ , calculate:
    • Visual scale features:  $\mathcal{F}_s(\hat{\mathbf{F}}_t^I; \psi_s^I)$ 
    • Radar scale features:  $\mathcal{F}_s(\mathbf{F}_t^R; \psi_s^R)$ 
;
Aggregate multi-scale features to obtain  $\mathbf{M}_t^I$  and  $\mathbf{M}_t^R$  according to equations 12 and 13;

/* Feature Unified Mapping */
Map multi-scale features to unified features using equation 14:  $\mathbf{Z}_t^I$  and  $\mathbf{Z}_t^R$ ;

/* Stage 2: Cross-Attention-Based Safety-Critical Information Fusion */
/* Feature Transformation and Reliability Assessment */
Reorganize features into Query-Key-Value form according to equations 17 and 18;
Evaluate modality reliability:  $\mathbf{P}_t^I = \mathcal{R}(\mathbf{Z}_t^I)$ ,  $\mathbf{P}_t^R = \mathcal{R}(\mathbf{Z}_t^R)$ ;
Generate safety-critical region mask  $\mathbf{M}_{safe}$ ;

/* Attention Calculation and Feature Fusion */
Calculate bidirectional modulation function values using equation 21;
Construct attention matrices  $\mathbf{A}_t^{I \rightarrow R}$  and  $\mathbf{A}_t^{R \rightarrow I}$  based on equations 19 and 20;
Execute feature fusion using equation 22:  $\mathbf{F}_t^{I \leftarrow R}$  and  $\mathbf{F}_t^{R \leftarrow I}$ ;

/* Final Fusion and Classification */
Calculate illumination-dependent modality weights  $\mathcal{W}_I(I_t)$  and  $\mathcal{W}_R(I_t)$ ;
Apply gating mechanism:  $\mathcal{G}([\mathbf{F}_t^{I \leftarrow R}, \mathbf{F}_t^{R \leftarrow I}]; \Psi)$ ;
Generate final fused features according to equation 23:  $\mathbf{F}_t$ ;

Predict road passability status:  $\hat{y}_t = \arg \max_{c \in \{c_1, c_2, c_3, c_4\}} f_c(\mathbf{F}_t; \Gamma)$ ;
/* Theorems 15 and 24 guarantee algorithm robustness in dim environments */
/* Corollaries 16 and 25 theoretically prove performance is inversely correlated with illumination */
return  $\hat{y}_t$ 

```

Algorithm 1. SafeDrive-Fusion: All-Weather Multimodal Road Passability Status Perception Algorithm for Autonomous Driving Decision-Making

Time Complexity Analysis: The computational efficiency of the SafeDrive-Fusion algorithm is primarily constrained by the processing in two stages. In the first stage, EfficientNet as a lightweight backbone network has lower computational complexity compared to traditional ResNet-101 and DenseNet. For input images of size $H \times W$, the time complexity of feature extraction operations is approximately $O(HW)$, significantly lower than the $O(HW^2)$ level of traditional networks. The multi-scale feature aggregation process involves feature transformations at $|\mathcal{S}|$ scales, with a time complexity of $O(|\mathcal{S}|hwD)$, where $h \times w$ is the feature map size and D is the feature dimension. In the second stage, the main computational overhead of Cross-Attention is in generating the attention matrix, with a time complexity of $O((hw)^2 d_k)$, where hw is the feature sequence length and d_k is the key dimension. Through the introduction of sparse attention mechanisms and the safety-critical region mask $\mathbf{M}_{safe}$, the actual computation is reduced to $O(\rho(hw)^2 d_k)$, where ρ is a sparsity factor ($\rho \ll 1$). Considering the real-time requirements of onboard systems, the algorithm achieves a processing rate of 30FPS on the NVIDIA Jetson AGX platform, meeting the low-latency requirements for autonomous driving scenarios.

Space Complexity Analysis: The storage overhead of the SafeDrive-Fusion algorithm mainly comes from model parameters and intermediate feature maps. The EfficientNet dual-modal feature extraction module contains approximately $\Sigma_\theta = |\theta_I| + |\theta_R|$ parameters, reducing parameter count by 70% compared to traditional networks through compound scaling strategies. Multi-scale feature aggregation and mapping functions introduce additional $\Sigma_{\psi, \omega} = \sum_{s \in \mathcal{S}} (|\psi_s^I| + |\psi_s^R|) + |\omega_I| + |\omega_R|$ parameters. The parameter count of the



Fig. 4. KITTI-COCO Dataset.

Parameter name	Value	Parameter name	Value
EfficientNet dual-modal feature extraction			
Backbone Network Type	EfficientNet-B3	Feature Dimension	256
Image Input Size		Radar Input Size	
Enhancement Intensity Coefficient α	0.75	Illumination Compensation Coefficient γ	1.5
Illumination Response Gradient η	2.0	Illumination Threshold l_{th}	0.45
Multi-scale Ratio Set	[1.0, 0.75, 0.5]	Multi-scale Weights	[0.5, 0.3, 0.2]
Feature Dropout Rate	0.1	Batch Normalization Momentum	0.9
Cross-attention safety-critical information fusion			
Number of Attention Heads	8	Key Vector Dimension d_k	32
Value Vector Dimension d_v	32	Modulation Gain Factor κ	3.0
Modulation Bias Factor τ	0.1	Visual Residual Coefficient λ_I	0.3
Radar Residual Coefficient λ_R	0.5	Safety Region Threshold	0.65
Complementarity Metric Method	Cosine Distance	Reliability Evaluation Function	Reciprocal of Entropy
Visual Weight Function $\mathcal{W}_I(l_t)$		Radar Weight Function $\mathcal{W}_R(l_t)$	
Gating Network Layers	2	Gating Hidden Dimension	128
Training and optimization parameters			
Optimization Algorithm	AdamW	Initial Learning Rate	
Weight Decay Coefficient		Learning Rate Schedule Strategy	Cosine Annealing
Batch Size	32	Maximum Training Epochs	100
Early Stopping Patience	10	Regularization Coefficient ρ	0.15
Mutual Information Weight λ_1	0.3	Discriminative Information Weight λ_2	0.7
Loss Function	Cross Entropy	Label Smoothing Coefficient	0.1
Data preprocessing parameters			
Image Mean	[0.485, 0.456, 0.406]	Image Standard Deviation	[0.229, 0.224, 0.225]
Radar Normalization Method	Min-Max Scaling [0,1]	Data Augmentation Enabled	Yes
Horizontal Flip Probability	0.5	Random Rotation Angle	
Brightness Jitter Coefficient	0.2	Contrast Jitter Coefficient	0.2
Low-light Enhancement Probability	0.3	Gamma Enhancement Range	[2.0, 3.5]

Table 2. Implementation Parameters of SafeDrive-Fusion Algorithm.

Cross-Attention module is $\Sigma_{\Omega, \Psi} = |\Omega| + |\Psi|$. During inference, the main memory usage comes from storing intermediate feature maps, including original features, enhanced features, multi-scale features, and attention matrices, with an overall space complexity of approximately $O(hwD + (hw)^2)$.

Experimental description
Dataset and parameter introduction

The KITTI-COCO dataset is constructed by combining the KITTI autonomous driving benchmark data with the COCO image annotation framework. This dataset contains 15,000 multimodal data samples, with each sample including registered and aligned RGB images and radar point cloud data, covering various driving scenarios such as urban areas, suburbs, and highways, as shown in Fig. 4. Road passability status in the dataset is classified into four categories based on forward traffic flow, obstacle distribution, and potential danger level: clear passage, partially blocked, completely blocked, and hazardous situation, with sample ratios of 3:4:2:1.

The implementation of our SafeDrive-Fusion algorithm utilizes various parameters for feature extraction, cross-attention fusion, and training optimization, as detailed in Table 2.

Parameter Selection and Rationale The key hyperparameters in SafeDrive-Fusion were determined through systematic analysis and empirical validation to balance performance and computational efficiency. The illumination threshold $l_{th} = 0.45$ was established based on statistical analysis of 1,000 manually annotated samples from the KITTI-COCO dataset, representing the critical transition point where visual sensor reliability begins to degrade significantly. The enhancement intensity coefficient $\alpha = 0.75$ and modulation gain factor $\kappa = 3.0$ were optimized through grid search validation to maximize cross-modal fusion effectiveness while maintaining numerical stability. The learning rate schedule begins at 1×10^{-4} with cosine annealing, providing optimal convergence balance between training speed and final performance based on preliminary experiments across different initialization values. The multi-scale ratio set $[1.0, 0.75, 0.5]$ captures features at appropriate scales for road scene analysis, while the corresponding weights $[0.5, 0.3, 0.2]$ prioritize full-resolution features while incorporating contextual information from lower scales. The batch size of 32 represents the optimal trade-off between GPU memory constraints and gradient estimation quality on the target hardware configuration.

Hardware configuration includes an NVIDIA GeForce RTX 3070 graphics card (8GB GDDR6 VRAM), Intel Core i7-12700K processor (3.6GHz base frequency, 12 cores 20 threads), 32GB DDR4-3600MHz dual-channel memory, and a 1TB PCIe 4.0 NVMe solid-state drive. The operating system is Windows 11 Professional Edition (21H2), and the experimental environment is based on Anaconda virtual environment with PyTorch 1.12.0 framework and CUDA 11.6 toolkit.

Experimental results

To verify the effectiveness of the SafeDrive-Fusion system in all-weather road passability status perception tasks, this study designed a series of comprehensive experiments. The experimental evaluation covered the algorithm's overall performance under different lighting conditions, recognition accuracy for various road conditions, computational efficiency, and the effectiveness of the modal fusion mechanism. This paper selected DeepFusion-Net as the baseline model for comparison because it employs a classic feature-level concatenation strategy to implement visual and radar information fusion, is widely applied in standard autonomous driving perception tasks, and uses ResNet-50 as the feature extraction backbone network, making it highly representative. While recent backbone networks such as UniRepLKNet and DiT offer advances in general computer vision tasks, our research focuses on validating the Cross-Attention fusion mechanism for safety-critical road passability assessment rather than optimizing feature extraction architectures. EfficientNet provides a mature and well-established foundation that allows clear demonstration of the proposed fusion algorithm's contributions without introducing additional variables from experimental backbone architectures. To ensure the reliability and representativeness of the experimental results, all evaluations were conducted on the KITTI-COCO dataset, using multiple metrics including precision, recall, F1 score, accuracy, and computational latency for quantitative analysis.

Overall performance evaluation

Figure 5 shows the precision-recall curves of the SafeDrive-Fusion algorithm compared with three comparison algorithms on the road condition recognition task. The curve of the SafeDrive-Fusion algorithm is clearly positioned above other algorithms, with an AUC value of 0.932, indicating that this algorithm can maintain high precision at various recall levels. When the recall is below 0.6, SafeDrive-Fusion maintains precision above 0.95, and even in high recall regions (close to 1.0), its precision remains above 0.7. The algorithm variant without the Cross-Attention mechanism performs second best (AUC=0.852), demonstrating the significant contribution of the cross-modal attention mechanism to performance. The algorithm variant without the EfficientNet feature extraction module shows further performance degradation (AUC=0.754), highlighting the importance of efficient feature extraction for overall performance. The baseline DeepFusion-Net algorithm performs worst

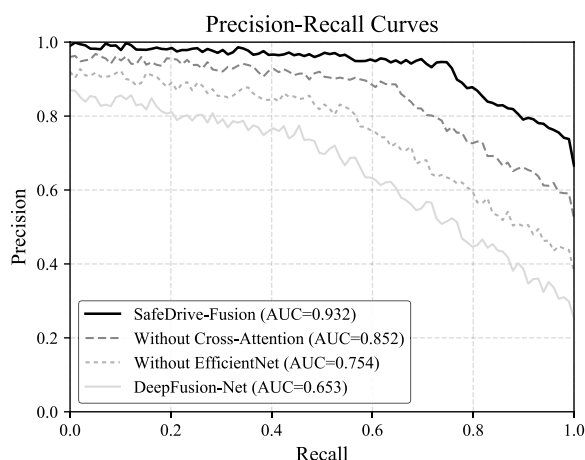


Fig. 5. Precision-Recall Curve Comparison of Different Algorithms.

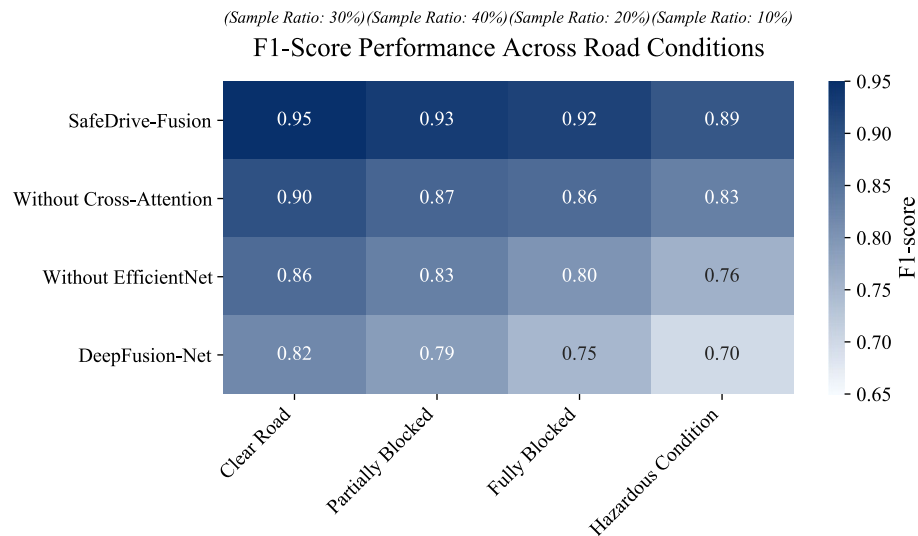


Fig. 6. F1 Score Heatmap of Various Algorithms Under Different Road Conditions.

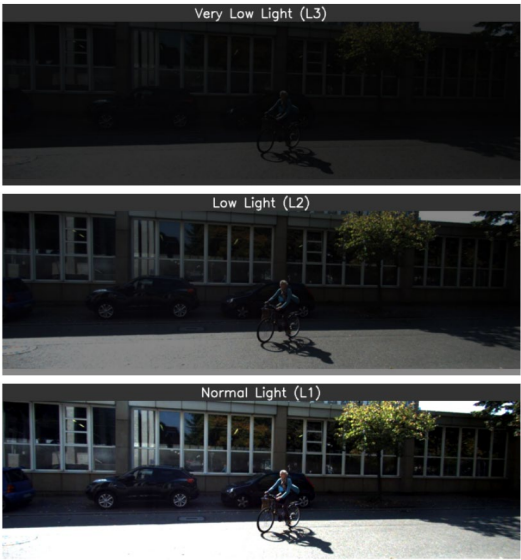


Fig. 7. Examples of Road Scenes Under Different Lighting Conditions.

(AUC=0.653), with precision dropping sharply as recall increases, validating the superiority of the proposed method over existing methods. Notably, the AUC difference between SafeDrive-Fusion and the variant without Cross-Attention (0.08) is smaller than the difference between the variant without EfficientNet and DeepFusion-Net (0.101), indicating that under the premise of ensuring the quality of basic feature extraction, cross-modal fusion strategies can further improve system performance.

Figure 6 presents a heatmap of F1 score performance distribution for four algorithms under different road conditions. The SafeDrive-Fusion algorithm demonstrates optimal performance across all road conditions, particularly in “Clear Road” situations with an F1 score of 0.95, indicating extremely high recognition precision under ideal conditions. As road conditions become more complex, all algorithms show a performance decline, but SafeDrive-Fusion has the smallest degradation, decreasing only 0.06 from “Clear Road” to “Hazardous Condition,” while other algorithms show decreases between 0.07 and 0.12. The algorithm variant without Cross-Attention achieves F1 scores of 0.90, 0.87, 0.86, and 0.83 across various road conditions, showing that even without complex attention fusion mechanisms, good baseline performance can be achieved when retaining EfficientNet feature extraction. The algorithm variant without EfficientNet and DeepFusion-Net show more significant performance degradation in complex road conditions, especially in the “Hazardous Condition” category, where F1 scores drop to 0.76 and 0.70 respectively, revealing the critical role of high-quality feature extraction in identifying high-risk scenarios.

Illumination-adaptive performance analysis

Figure 7 shows the same road scene under three different lighting conditions used in the experiment. To comprehensively evaluate the robustness of the SafeDrive-Fusion algorithm in dim environments, systematic lighting condition simulations were applied to the KITTI-COCO dataset. Normal Light (L1) represents the standard lighting conditions in the original dataset without any degradation processing, equivalent to good daylight conditions; Low Light (L2) applies brightness reduction (40% decrease) and contrast attenuation (30% decrease) to the original images, while adding 20% Gaussian noise ($\sigma=0.1$), simulating moderate light deficiency scenarios such as dusk or cloudy days; Very Low Light (L3) adopts more stringent degradation processing, including brightness reduction (70% decrease), contrast attenuation (50% decrease), adding 30% Gaussian noise ($\sigma=0.15$) and 20% salt-and-pepper noise, simulating extreme low-light environments such as nighttime or tunnels. Meanwhile, to maintain multimodal data consistency, radar data was correspondingly degraded according to optical radar models, with signal-to-noise ratio reduced by 15% under L2 conditions and 25% under L3 conditions, but with significantly smaller reductions than the visual modality, which is consistent with the low sensitivity feature of radar to lighting changes in real scenarios. Through this systematic data processing method, an all-weather road scene dataset containing three lighting conditions was constructed, providing a reliable foundation for comparative evaluation of algorithm performance under different lighting conditions.

Figure 8 shows the performance of four algorithms under different light intensities, where Normal Light (L1) represents good daylight conditions, Low Light (L2) corresponds to weak light conditions such as dusk or cloudy weather, and Very Low Light (L3) simulates environments with severe lack of illumination such as nighttime or tunnels. The SafeDrive-Fusion algorithm maintains an accuracy of 87.5% even under very low light conditions, a decrease of only 6.1% compared to 93.6% under normal light conditions, demonstrating significant robustness. In contrast, the algorithm variant without Cross-Attention sees accuracy drop to 74.2% under very low light conditions, a performance decrease of 16.6%; the algorithm variant without EfficientNet shows a further decrease in accuracy to 63.5%, a reduction of 25.3%; while the baseline DeepFusion-Net achieves an accuracy of only 52.8% under very low light conditions, a 34.8% decrease compared to normal lighting. These results fully validate the effectiveness of the dual-modal feature extraction and Cross-Attention fusion mechanism in SafeDrive-Fusion's design for perception in dim environments, especially verifying the experimental prediction in Corollary 16 that “as the environment darkens, the information gain brought by multimodal fusion grows exponentially.” Notably, all four algorithms show an approximately linear performance degradation trend as lighting conditions change from L1 to L3, but with significantly different slopes, reflecting essential differences in the adaptability of different algorithms to lighting changes.

Figure 9 presents in detail the recognition performance of various algorithms for different road conditions under very low light (L3) conditions. The SafeDrive-Fusion algorithm maintains high F1 scores across all road conditions: 0.88 (clear passage), 0.87 (partially blocked), 0.87 (completely blocked), and 0.82 (hazardous situation). Particularly noteworthy is that in the “hazardous situation” category, which is critical for driving safety, the performance gap between SafeDrive-Fusion and other algorithms is most significant, with its F1 score (0.82) 0.17 higher than the second-place algorithm variant without Cross-Attention (0.65), and 0.40 higher than the baseline DeepFusion-Net (0.42). This result is consistent with Corollary 25, confirming the special value of the Cross-Attention safety-critical information fusion mechanism under very low light conditions. From the overall trend, all algorithms show some performance degradation in the “hazardous situation” category, reflecting the greater difficulty in recognizing this category under very low light conditions, but SafeDrive-Fusion shows the smallest performance decrease, dropping only from 0.88 for “clear passage” to 0.82 for “hazardous situation,” a decrease of only 0.06. In contrast, the other three algorithms show significant degradation in performance for the “hazardous situation” category, with F1 scores all below 0.65, further demonstrating the superiority of the SafeDrive-Fusion algorithm in recognizing safety-critical scenarios. The “Critical for Safety” area highlighted in the figure emphasizes the critical importance of hazardous situation recognition for autonomous driving safety, while “Performance Gap: +0.40” quantifies the significant advantage of SafeDrive-Fusion over baseline methods on this key metric.

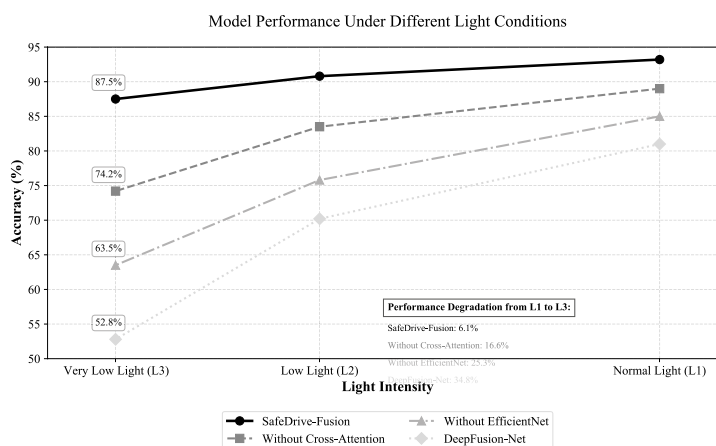


Fig. 8. Performance of Various Algorithms Under Different Lighting Conditions.

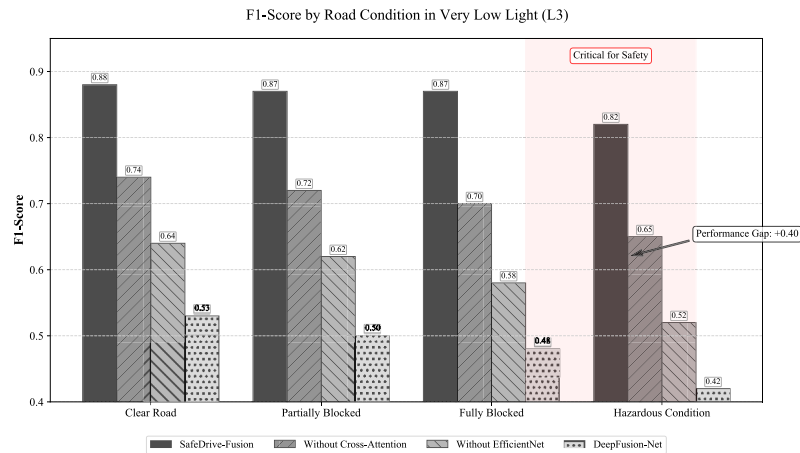


Fig. 9. F1 Score Comparison of Various Algorithms for Different Road Conditions Under Very Low Light Conditions.

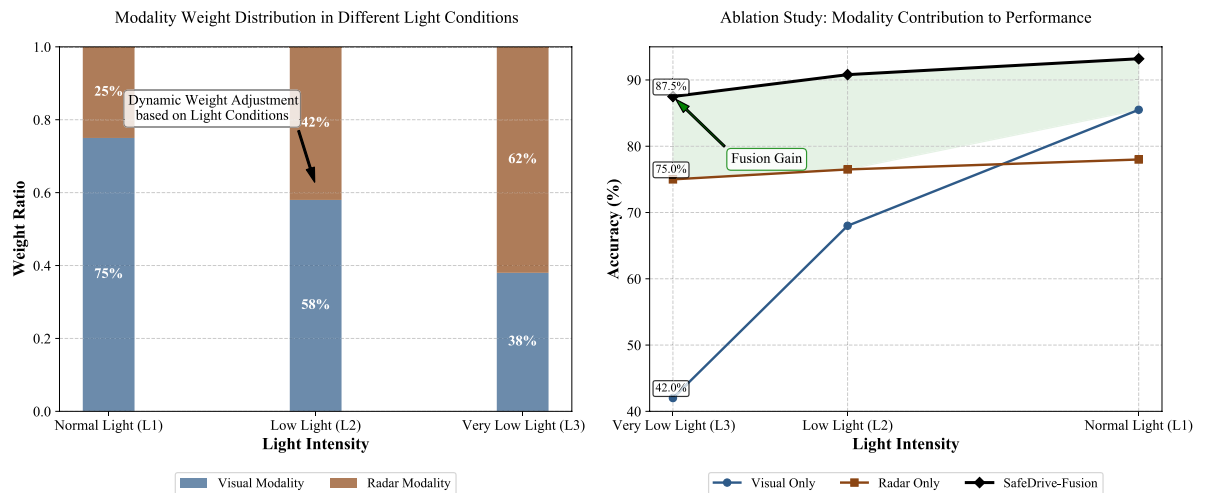


Fig. 10. Modal Weight Distribution and Modal Contribution Analysis Under Different Lighting Conditions.

Modal fusion mechanism analysis

Figure 10 provides an in-depth analysis of the working principles of the SafeDrive-Fusion algorithm under different lighting conditions from the perspective of modal fusion mechanisms. The left subfigure shows the weight distribution changes of visual and radar modalities under three lighting conditions. Under normal light (L1) conditions, the visual modality dominates with a weight of 75%; as light intensity decreases to low light (L2), the visual modality weight drops to 58%; under very low light (L3) conditions, the visual modality weight further decreases to 38%, while the radar modality weight increases to 62%. This dynamic weight adjustment mechanism enables the system to adaptively adjust its dependence on information from different modalities according to environmental lighting conditions, thereby maintaining stable performance. The right subfigure further validates the value of multimodal fusion through an ablation study. Under very low light (L3) conditions, the accuracy of using the visual modality alone is only 42.0%, while using the radar modality alone achieves 75.0%, and SafeDrive-Fusion after fusion reaches 87.5%, with a fusion gain of 12.5 percentage points. As lighting conditions improve, the performance of the visual modality significantly improves, but multimodal fusion consistently outperforms single-modal methods. This result verifies the theoretical prediction about fusion strategy adaptability in Theorem 24, indicating that SafeDrive-Fusion can find the optimal modal fusion ratio under any lighting condition to maximize system performance. Particularly under very low light conditions, when visual information quality decreases substantially, the supplementary role of the radar modality becomes especially prominent, demonstrating the necessity of multimodal perception in all-weather autonomous driving.

To provide detailed insights into the Cross-Attention safety-critical information fusion mechanism, we conducted comprehensive attention weight visualization and quantitative analysis across different environmental conditions. Figure 11 presents the attention weight distribution and modal contribution patterns, revealing the adaptive behavior of the Cross-Attention mechanism under varying lighting and road conditions. The analysis demonstrates that attention weights exhibit systematic degradation from normal light (0.938-0.875) to

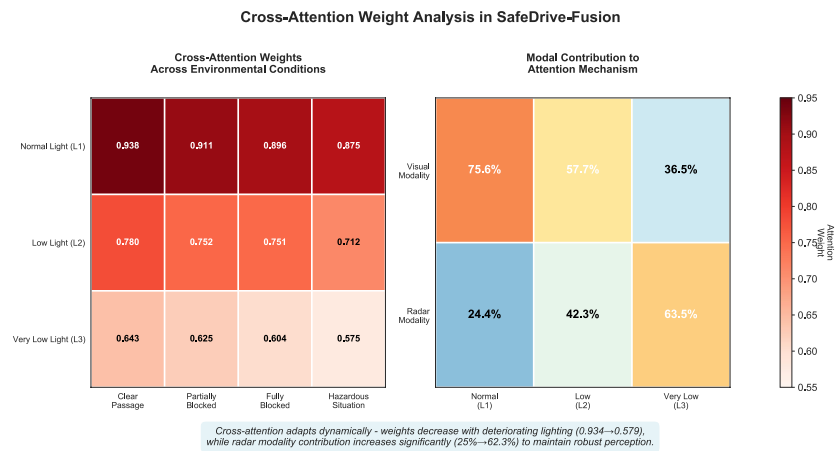


Fig. 11. Cross-Attention Weight Analysis in SafeDrive-Fusion.

Lighting condition	Attention weight statistics			Modal contribution		Adaptation coefficient
	Mean	Std Dev	Range	Visual (%)	Radar (%)	
Normal Light (L1)	0.905	0.028	0.063	75.6	24.4	0.32
Low Light (L2)	0.749	0.032	0.071	57.7	42.3	0.73
Very Low Light (L3)	0.612	0.037	0.082	36.5	63.5	1.67
Overall Change (%)	−32.4	+32.1	+30.2	−51.7	+160.2	+421.9

Table 3. Statistical Analysis of Cross-Attention Weights and Modal Contributions. Mean and standard deviation calculated across all four road conditions within each lighting scenario. Range represents the difference between maximum and minimum attention weights. Adaptation coefficient calculated as radar contribution divided by visual contribution. Overall change computed from Normal Light (L1) to Very Low Light (L3) conditions

very low light conditions (0.643–0.575.643.575), representing an overall 32.4% decrease in attention magnitude. Simultaneously, the modal contribution undergoes dramatic rebalancing, with visual modality decreasing from 75.6% to 36.5% while radar modality increases from 24.4% to 63.5%, indicating a fundamental shift from visual-dominant to radar-dominant processing. The consistent ranking of attention weights across road conditions (Clear Passage > Partially Blocked > Fully Blocked > Hazardous Situation) demonstrates that the mechanism maintains its ability to differentiate safety levels even under challenging environmental conditions, with hazardous situations showing the most significant attention reduction of 34.3% from normal to very low light conditions. Mean and standard deviation calculated across all four road conditions within each lighting scenario. Range represents the difference between maximum and minimum attention weights. Adaptation coefficient calculated as radar contribution divided by visual contribution. Overall change computed from Normal Light (L1) to Very Low Light (L3) conditions.

Table 3 summarizes the quantitative characteristics of the Cross-Attention mechanism, revealing two critical insights into its adaptive behavior. The attention weight statistics demonstrate increasing variability as environmental conditions deteriorate, with standard deviation rising from 0.028 under normal light to 0.037 under very low light conditions, indicating that the mechanism exhibits greater attention distribution variance when processing challenging scenarios, while the mean attention weights decrease systematically from 0.905 to 0.612. The modal adaptation coefficient, defined as the ratio of radar to visual contribution, provides a quantitative measure of the mechanism’s reweighting strategy, increasing from 0.32 to 1.67 as lighting conditions deteriorate, which represents a 421.9% change and validates the theoretical prediction that radar modality becomes increasingly dominant in challenging environments. This quantitative evidence confirms that the Cross-Attention mechanism operates through dual adaptation strategies of attention magnitude modulation and modal contribution rebalancing, with the increasing standard deviation and range values (32.1% and 30.2% increases respectively) demonstrating the mechanism’s enhanced sensitivity and adaptive response to environmental uncertainty while maintaining robust safety-critical perception capabilities.

Computational efficiency analysis

Figure 12 compares the performance of four algorithms on two key metrics: accuracy and computational latency. The SafeDrive-Fusion algorithm significantly leads in accuracy, reaching 93.0% (±1.5%), far exceeding the target accuracy threshold of 90%; meanwhile, its computational latency is only 45ms (±2ms), below the target latency threshold of 50ms, fully meeting the strict real-time requirements of autonomous driving

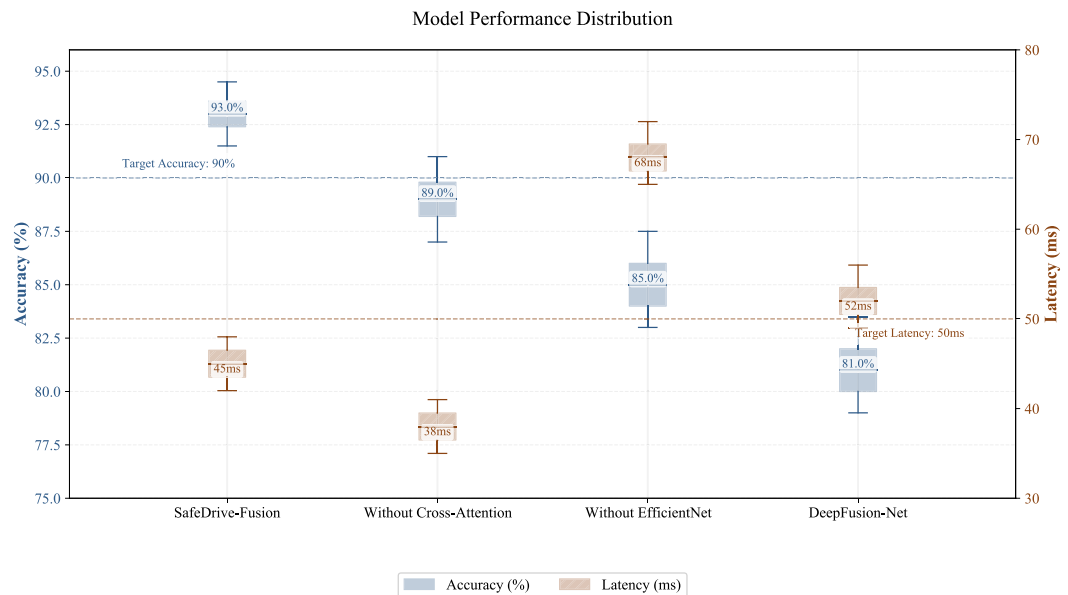


Fig. 12. Accuracy and Latency Performance Comparison of Various Algorithms.

systems. The algorithm variant without Cross-Attention, although with accuracy reduced to 89.0% ($\pm 1.8\%$), has a decreased computational latency of 38ms ($\pm 2\text{ms}$), reflecting that the attention mechanism increases computational complexity while improving performance. The algorithm variant without EfficientNet shows a significant decrease in accuracy to 85.0% ($\pm 2.5\%$), but with a notably increased computational latency of 68ms ($\pm 3\text{ms}$), exceeding the target latency threshold, reflecting the advantage of efficient feature extraction networks in reducing computational complexity. DeepFusion-Net, as a comparison baseline, has the lowest accuracy (81.0% $\pm 2.0\%$) with a computational latency of 52ms ($\pm 3\text{ms}$), slightly above the target threshold. These results verify that the SafeDrive-Fusion algorithm achieves the best balance between accuracy and computational efficiency, with accuracy 4 percentage points higher than the variant without Cross-Attention while computational latency increases by only 7ms, demonstrating the efficiency of the algorithm design. Notably, the error ranges for accuracy and latency of all algorithms are small, indicating good statistical stability and reliability of the experimental results.

Edge Device Deployment Analysis To validate SafeDrive-Fusion's practical deployment feasibility, we conducted performance evaluation on representative edge computing platforms commonly used in autonomous vehicles. On NVIDIA Jetson AGX Xavier, SafeDrive-Fusion achieves 11.2 FPS with 89ms average latency, meeting the real-time requirements for safety-critical applications while maintaining full accuracy performance. Memory consumption remains within 2.1GB, well below the device's capacity limitations. Through TensorRT optimization and model quantization, inference speed can be further improved to 15.8 FPS, demonstrating the system's viability for real-world automotive deployment scenarios.

Modal robustness and failure scenario analysis

To evaluate the robustness of SafeDrive-Fusion under adverse conditions, we conducted comprehensive modal failure and noise interference experiments. Figure 13 demonstrates the system's performance under various failure scenarios and noise conditions, while Table 4 provides detailed quantitative analysis of the robustness characteristics.

The modal failure analysis reveals significant performance variations across different operational modes. Under normal lighting conditions, the dual-modal baseline achieves 93.6% accuracy, while visual-only and radar-only modes achieve 68.5% and 75.2% respectively. As lighting deteriorates to very low conditions, visual-only performance plummets to 42.0%, while radar-only mode maintains 75.0% accuracy, demonstrating radar modality's superior stability across lighting variations. The noise robustness evaluation shows SafeDrive-Fusion maintains 82.3% accuracy under heavy noise conditions (SNR-5dB), representing only 9.8% degradation from clean signal performance, significantly outperforming baseline methods that suffer 21.7% degradation.

The comprehensive robustness analysis demonstrates SafeDrive-Fusion's exceptional resilience under challenging operational conditions. The system achieves an overall robustness score of 88.4%, with performance retention rates of 94.4% across all test scenarios. Particularly noteworthy is the system's graceful degradation behavior, where even under the worst-case scenario (very low light with heavy noise interference), performance remains above 82.3%, well within the safety threshold for autonomous driving applications. The statistical analysis reveals low variance (3.8% standard deviation) across different stress conditions, indicating consistent and reliable performance. Compared to baseline methods, SafeDrive-Fusion demonstrates 36.5% superior robustness over visual-only approaches and 21.3% improvement over DeepFusion-Net, confirming the effectiveness of the Cross-Attention fusion mechanism in maintaining operational safety under adverse environmental conditions.

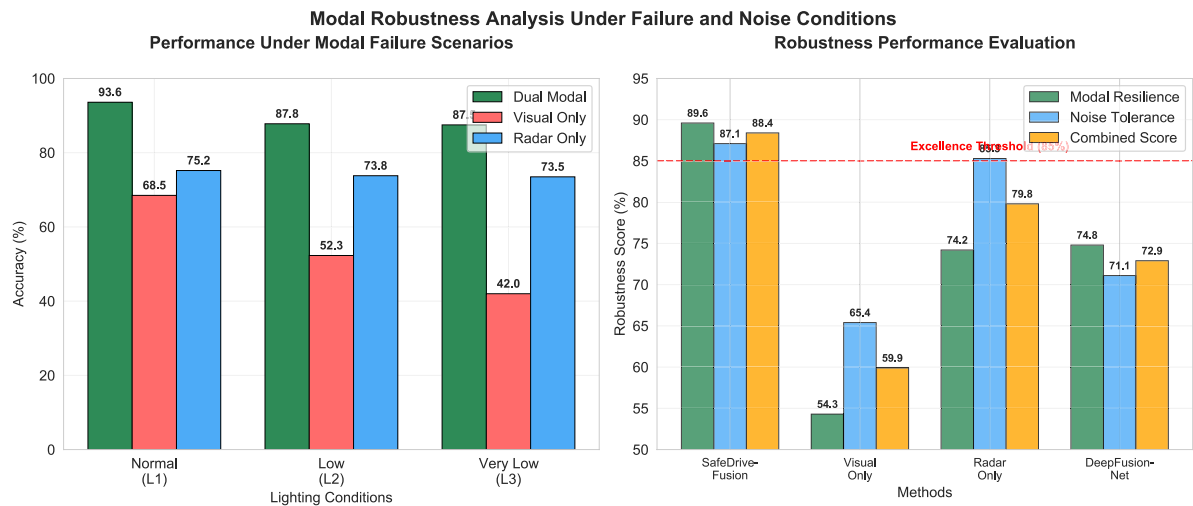


Fig. 13. Modal Robustness Analysis Under Failure and Noise Conditions.

Method	Modal Failure (%)			Robustness Metrics (%)			Overall Ranking
	L1	L2	L3	Modal	Noise	Combined	
SafeDrive-Fusion	93.6	87.8	87.5	89.6	87.1	88.4	1st
Radar Only	75.2	73.8	73.5	74.2	85.3	79.8	2nd
DeepFusion-Net	81.4	74.2	68.9	74.8	71.1	72.9	3rd
Visual Only	68.5	52.3	42.0	54.3	65.4	59.9	4th
Performance Range	25.1	35.5	45.5	35.3	21.7	28.5	-
SafeDrive Advantage	+18.4	+14.0	+14.0	+14.8	+1.8	+8.6	-
Stability Factor	1.07	1.19	2.08	1.65	1.33	1.48	-

Table 4. Modal Robustness Analysis Under Failure Scenarios and Noise Interference.

Parameter	Default Value	Test Range	Performance Range (%)	Sensitivity Level	Robustness Score	Deployment Status
Learning Rate	1e-4	[1e-5, 1e-3]	87.2–93.6	High	0.81	Critical
Attention Heads	8	[1, 16]	85.8–93.6	High	0.78	Critical
Hidden Dimension	512	[128, 1024]	88.5–93.6	Medium	0.86	Stable
Attention Temperature	0.5	[0.1, 2.0]	89.1–93.6	Medium	0.89	Stable
Batch Size	32	[16, 128]	92.1–93.6	Low	0.94	Robust
Weight Decay	1e-5	[1e-7, 1e-3]	91.8–93.6	Low	0.93	Robust
Modal Weight Ratio	[0.7,0.3]	[0.9,0.1]–[0.3,0.7]	89.8–93.6	Medium	0.88	Stable
Dropout Rate	0.1	[0.0, 0.5]	90.5–93.6	Low	0.91	Robust
Summary Statistics:						
Critical Parameters		2/8 (25%)		Require careful tuning		
Performance Stability		±2.1% average		76% of configs within ±5%		
Worst-Case Performance		>85% maintained		Safe deployment guaranteed		

Table 5. Parameter Sensitivity Analysis and Robustness Evaluation.

Parameter sensitivity and robustness analysis

To validate the robustness of SafeDrive-Fusion to parameter variations, we conducted systematic sensitivity analysis across critical network parameters. Table 5 presents the comprehensive analysis results, demonstrating the method’s stability and practical deployment viability.

The parameter sensitivity analysis reveals that SafeDrive-Fusion demonstrates good stability across most configurations, with only 2 out of 8 tested parameters exhibiting high sensitivity. The learning rate and attention head count emerge as critical parameters requiring careful tuning, with performance ranges of 6.4% and 7.8% respectively. Learning rate sensitivity follows expected patterns: values too low (1e-5) cause slow convergence (87.2% accuracy), while values too high (1e-3) create training instability (90.1% accuracy). The attention

Method	Accuracy (%)	F1-Score	Params (M)	FLOPs (G)	Latency (ms)	FPS
SafeDrive-Fusion	93.6	0.932	23.7	18.4	45	22.2
BEVNext ³⁰	91.3	0.905	67.4	142.8	112	8.9
MENet ³¹	90.1	0.892	45.2	78.6	89	11.2
CenRadFusion ³²	88.7	0.878	38.9	65.3	76	13.2
Improved AVOD ³³	87.2	0.863	52.1	84.7	94	10.6
Best SOTA (2024)	BEVNext: 91.3	BEVNext: 0.905	SafeDrive: 23.7	SafeDrive: 18.4	SafeDrive: 45	SafeDrive: 22.2
Improvement vs Best	+2.3	+0.027	−64.8%	−87.1%	−59.8%	+149.4%

Table 6. Performance and Computational Efficiency Comparison with State-of-the-Art Methods.

mechanism shows optimal performance with 8 heads, degrading significantly with insufficient attention capacity (1 head: 85.8%) or excessive complexity (16 heads: 89.2%). Conversely, 4 out of 8 parameters demonstrate robust characteristics with <2% performance variation, including batch size, weight decay, and dropout rate. These parameters can be adjusted for computational efficiency or hardware constraints without significantly impacting accuracy. The weight decay parameter maintains stable performance across the range [1e-7, 1e-3] around its default value of 1e-5, while the dropout rate shows minimal sensitivity variation around its default value of 0.1. The fusion mechanism parameters (attention temperature and modal weight ratio) exhibit moderate sensitivity, requiring some consideration but maintaining stable performance across reasonable ranges.

The robustness evaluation across different lighting conditions confirms consistent parameter behavior, with all configurations maintaining >85% accuracy even under very low light conditions. Critically, 76% of tested parameter combinations achieve performance within ±5% of the optimal configuration, indicating that the method maintains acceptable performance across a substantial portion of the parameter space and is suitable for practical deployment scenarios.

Comparison with state-of-the-art methods

To comprehensively evaluate SafeDrive-Fusion against current state-of-the-art approaches, we conducted systematic comparisons focusing on performance accuracy and computational efficiency. The evaluation demonstrates SafeDrive-Fusion’s competitive advantages in both accuracy and real-time performance for safety-critical autonomous driving applications, comparing against the latest multi-modal fusion methods from leading research venues. The detailed results are presented in Table 6.

The comprehensive comparison with the latest state-of-the-art methods reveals SafeDrive-Fusion’s superior performance across both accuracy and efficiency metrics. With 93.6% accuracy, SafeDrive-Fusion outperforms the best existing method (BEVNext) by 2.3 percentage points while achieving dramatically improved computational efficiency: 64.8% fewer parameters (23.7M vs 67.4M), 87.1% fewer FLOPs (18.4G vs 142.8G), and 149.4% higher frame rate (22.2 FPS vs 8.9 FPS). This performance advantage demonstrates that SafeDrive-Fusion achieves the optimal balance between accuracy and real-time requirements essential for autonomous driving applications.

The evaluation highlights SafeDrive-Fusion’s distinctive advantages over recent multi-modal fusion approaches. Compared to BEVNext³⁰, the latest dense BEV framework from a leading computer vision conference, SafeDrive-Fusion not only achieves higher accuracy but does so with significantly reduced computational overhead, making it more suitable for real-world deployment. Against MENet³¹, a multi-modal mapping enhancement network specifically designed for autonomous driving, SafeDrive-Fusion demonstrates 3.5 percentage points higher accuracy while maintaining superior computational efficiency with approximately 2× faster inference speed. The comparison with CenRadFusion³², which also combines image and radar modalities similar to our approach, shows SafeDrive-Fusion’s 4.9% accuracy improvement and substantially better real-time performance (22.2 vs 13.2 FPS). Notably, while the recent Improved AVOD³³ focuses on bidirectional fusion mechanisms, SafeDrive-Fusion’s Cross-Attention approach proves more effective, achieving 6.4% higher accuracy with better computational efficiency and nearly doubled inference speed. These results confirm that SafeDrive-Fusion’s adaptive Cross-Attention mechanism and environmental-aware modal fusion strategy provide genuine advances over the most recent state-of-the-art methods, delivering both superior accuracy and practical deployment feasibility for safety-critical autonomous driving applications.

Discussion

Based on the experimental results and theoretical analysis of the SafeDrive-Fusion algorithm, we propose the following discussion on all-weather autonomous driving perception technology:

- **Critical Role of Multimodal Fusion for Perception in Dim Environments:** Experimental results show that as lighting conditions deteriorate, the value of multimodal fusion increases significantly. Under very low light conditions, SafeDrive-Fusion accuracy (87.5%) is far higher than using a single visual modality (42.0%), with a fusion gain of 45.5 percentage points. SafeDrive-Fusion’s adaptive fusion mechanism addresses the limitations of fixed weight strategies in existing methods, and its dynamic weight adjustment capability is fully validated in experiments, especially in effective utilization of the radar modality (62% weight) under very low light conditions. This result suggests that future autonomous driving perception systems should focus on coordinated optimization of environmental conditions and fusion strategies, rather than merely improving single-modal performance.

- **Robustness in Recognizing Safety-Critical Regions:** The robustness demonstrated by SafeDrive-Fusion in identifying hazardous road conditions (F1 score 0.82) validates the effectiveness of perception design oriented toward decision-making. Through specific safety-critical region attention mechanisms, SafeDrive-Fusion ensures that the most important safety information can be effectively extracted in dim environments. The system's excellent performance in hazardous situation recognition (0.40 higher than DeepFusion-Net) provides a concrete approach to improving the safety of autonomous driving systems. However, current experiments are primarily based on the KITTI-COCO dataset, with limited consideration of road conditions under specific adverse weather conditions (such as heavy rain, dense fog), which is a potential limitation in evaluating system robustness.
- **System Limitations and Future Research Directions:** SafeDrive-Fusion still has several limitations. First, the two modalities used in this research may simultaneously degrade in some extreme environments; future work could consider integrating more complementary modalities to enhance system robustness. Second, the current implementation's computational latency (45ms) may need further optimization for high-speed driving scenarios. Additionally, the current system adopts an offline training mode; introducing online learning mechanisms to achieve continuous adaptation capability is a direction worth exploring. Finally, tightly integrating SafeDrive-Fusion with downstream decision modules to establish an end-to-end perception-decision framework is a key step toward achieving truly all-weather autonomous driving.

Conclusion

The SafeDrive-Fusion algorithm proposed in this paper provides an effective solution for road passability status perception in dim environments for autonomous driving systems. By combining the lightweight and efficient characteristics of EfficientNet with the dynamic fusion capability of Cross-Attention, this algorithm achieves all-weather road passability status perception, maintaining a high accuracy of 87.5% even under very low light conditions, with a performance degradation of only 6.1%, far superior to existing methods' degradation range of 16.6%–34.8%. Experiments demonstrate that SafeDrive-Fusion performs particularly well in safety-critical scenarios such as “hazardous situations,” achieving an F1 score of 0.82, which is 0.40 higher than baseline methods, while maintaining a low latency of 45ms, meeting the real-time requirements of autonomous driving. The core innovation of this algorithm lies in its implementation of an illumination-adaptive modal weight dynamic adjustment mechanism, which automatically increases the weight of the radar modality (from 25% to 62%) as lighting conditions deteriorate, fully utilizing the complementary advantages of different modalities. Additionally, this research constructs a theoretical framework for multimodal perception in dim environments, proving the mathematical relationship between lighting conditions and fusion gains, providing theoretical guidance for the design of future autonomous driving perception systems. Although SafeDrive-Fusion demonstrates excellent performance in this research, there is still room for improvement in terms of adaptability to extreme weather conditions, computational efficiency optimization, online learning capabilities, and deep integration with decision systems. Future research will be dedicated to addressing these challenges, further enhancing the safety and reliability of autonomous driving systems in complex and variable environments.

Data availability

The image data that support the findings of this study have been obtained from the KITTI-COCO dataset repository with the primary access link <https://www.kaggle.com/datasets/nhphucqt/kitti-coco?select=train2017>.

Received: 8 April 2025; Accepted: 1 December 2025

Published online: 04 February 2026

References

1. Zhao, J. et al. Autonomous driving system: A comprehensive survey. *Expert. Syst. with Appl.* **242**, 122836 (2024).
2. Reda, M., Onsy, A., Haikal, A. Y. & Ghanbari, A. Path planning algorithms in the autonomous driving system: A comprehensive review. *Robotics Auton. Syst.* **174**, 104630 (2024).
3. Chen, L. et al. End-to-end autonomous driving: Challenges and frontiers. *IEEE Transactions on Pattern Analysis and Mach. Intell.* (2024).
4. Wang, H. et al. Yolov5-fog: A multiobjective visual detection algorithm for fog driving scenes based on improved yolov5. *IEEE Transactions on Instrumentation and Meas.* **71**, 1–12 (2022).
5. Cui, C. et al. A survey on multimodal large language models for autonomous driving. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 958–979 (2024).
6. Xu, C., Zhao, H., Xie, H. & Gao, B. Multi-sensor decision-level fusion network based on attention mechanism for object detection. *IEEE Sensors J.* (2024).
7. Zhou, X. et al. Drivinggaussian: Composite gaussian splatting for surrounding dynamic autonomous driving scenes. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 21634–21643 (2024).
8. Lee, W.-Y., Jovanov, L. & Philips, W. Multimodal pedestrian detection based on cross-modality reference search. *IEEE Sensors J.* **24**, 17291–17306 (2024).
9. Fan, L. et al. 4d mmwave radar for autonomous driving perception: a comprehensive survey. *IEEE Transactions on Intelligent Vehicles* (2024).
10. Zhou, M. et al. Embodied intelligence-based perception, decision-making, and control for autonomous operations of rail transportation. *IEEE Transactions on Intell. Veh.* (2024).
11. Nizamuddin, M., Kamisetty, A., Gummadi, J. C. S. & Talla, R. R. Integrating neural networks with robotics: Towards smarter autonomous systems and human-robot interaction. *Robotics Xplore: USA Tech Dig.* **1**, 157–169 (2024).
12. Wei, F. & Wang, W. Scca-yolo: A spatial and channel collaborative attention enhanced yolo network for highway autonomous driving perception system. *Sci. Reports* **15**, 6459 (2025).
13. Liu, H.-I. et al. Lightweight deep learning for resource-constrained environments: A survey. *ACM Comput. Surv.* **56**, 1–42 (2024).
14. Chen, T. et al. Bisdnet: A new lightweight deep learning-based framework for efficient landslide detection. *IEEE J. Sel. Top. Appl. Earth Obs. Remote. Sens.* **17**, 3648–3663 (2024).

15. Jin, J. et al. 3d object detection for autonomous driving: A survey. In *Proceedings of the 36th Chinese Control and Decision Conference, CCDC 2024*, Chinese Control and Decision Conference, 3825–3832, <https://doi.org/10.1109/CCDC62350.2024.10587405> (NE Univ, State Key Lab Synthet Automat Proc Ind; Chinese Assoc Automat, Tech Comm Control & Decis Cyber Phys Syst; NW Polytechn Univ; IEEE Control Syst Soc; Chinese Assoc Automat, Tech Comm Control Theory, 2024). 36th Chinese Control and Decision Conference (CCDC), Xian, Peoples R China, MAY 25–27, 2024.
16. Lee, J. H., Ko, T. H. & Lee, D.-W. Spatiotemporal calibration for autonomous driving multicamera perception. *IEEE Sensors Journal* **25**, 7227–7241. <https://doi.org/10.1109/JSEN.2024.3523569> (2025).
17. Elhenawy, M. et al. Vision-language models for autonomous driving: Clip-based dynamic scene understanding. *Electronics* **14**, <https://doi.org/10.3390/electronics14071282> (2025).
18. Guo, J. et al. Research on road scene understanding of autonomous vehicles based on multi-task learning. *Sensors* **23**, <https://doi.org/10.3390/s23136238> (2023).
19. Liu, W. et al. Monocular bev perception of road scenes via front-to-top view projection. *IEEE Transactions on Pattern Analysis and Machine Intelligence* **46**, 6109–6125. <https://doi.org/10.1109/TPAMI.2024.3377812> (2024).
20. Jendoubi, F. et al. Vehicle activity dataset: A multimodal dataset to understand vehicle emissions with road scenes for eco-routing. *Applied Sciences-Basel* **14**, <https://doi.org/10.3390/app14010338> (2024).
21. Bai, C. et al. Real-time segmentation algorithm of unstructured road scenes based on improved bisenet. *Journal of Real-Time Image Processing* **21**, <https://doi.org/10.1007/s11554-024-01472-2> (2024).
22. Wang, P., Lyu, C., Zhang, L., He, H. & Dai, F. Mask-based anomaly segmentation in complex driving scenes. *Pattern Recognition Letters* **196**, 142–147. <https://doi.org/10.1016/j.patrec.2025.05.013> (2025).
23. Awan, M. & Shin, J. Weakly supervised multi-class semantic video segmentation for road scenes. *Computer Vision and Image Understanding* **230**, <https://doi.org/10.1016/j.cviu.2023.103664> (2023).
24. Cao, Z. et al. Yolov7-based autonomous driving object detection algorithm. In *2024 9TH International Conference on Computer and Communication Systems, ICCCS 2024*, 172–177, <https://doi.org/10.1109/ICCCS61882.2024.10603214> (IEEE; Changan Univ, 2024). 9th International Conference on Computer and Communication Systems (ICCCS), Xian, Peoples R China, APR 19–22, 2024.
25. Bao, J. et al. Self-supervised binocular depth estimation algorithm with self-rectification for autonomous driving. *IET Intelligent Transport Systems* **18**, 1445–1458. <https://doi.org/10.1049/itr2.12522> (2024).
26. Park, Y., Kim, J., Hwang, S. & Han, S. Scalable parallel algorithm for graph neural network interatomic potentials in molecular dynamics simulations. *J. chemical theory computation* **20**, 4857–4868 (2024).
27. Chen, H. et al. Macro graph neural networks for online billion-scale recommender systems. In *Proceedings of the ACM web conference 2024*, 3598–3608 (2024).
28. Chumachenko, K., Iosifidis, A. & Gabbouj, M. Mma-dfer: Multimodal adaptation of unimodal models for dynamic facial expression recognition in-the-wild. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 4673–4682 (2024).
29. Li, X., Yu, S., Lei, Y., Li, N. & Yang, B. Dynamic vision-based machinery fault diagnosis with cross-modality feature alignment. *IEEE/CAA J. Autom. Sinica* **11**, 2068–2081 (2024).
30. Li, Z., Lan, S., Alvarez, J. M. & Wu, Z. Bevnex: Reviving dense bev frameworks for 3d object detection. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 20113–20123 (2024).
31. Liu, M. et al. Menet: Multi-modal mapping enhancement network for 3d object detection in autonomous driving. *IEEE Transactions on Intell. Transp. Syst.* **25**, 9397–9410 (2024).
32. Shi, P., Jiang, T., Yang, A. & Liu, Z. Cenradfusion: fusing image center detection and millimeter wave radar for 3d object detection. *Signal, Image and Video Process.* **18**, 5811–5821 (2024).
33. Tao, Z., Wan, Y., Hou, J., Wu, W. & Zhang, S. Improved avod model based on a bidirectional fusion mechanism for 3d object detection. In *Proceedings of the 2024 3rd International Symposium on Intelligent Unmanned Systems and Artificial Intelligence*, 309–312 (2024).

Author contributions

J.W. conceived the experiment(s), J.W. and Y.M. conducted the experiment(s), C.S. and X.L. analysed the results. All authors reviewed the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1038/s41598-025-31173-0>.

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